

Shahriar Mahmud Sadee

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in Shahriar Mahmud Sadee 🌐 Shahriar Mahmud Sadee

Education

Chittagong University of Engineering and Technology

B.Sc. in Electrical and Electronic Engineering

February 2020 – July 2025

- GPA: 3.21/4.0

Relevant Courseworks

- Electromagnetic Field
- Microwave and Antenna Engineering
- Telecommunication Engineering
- Advance Communication Engineering
- Communication Network
- Electrical Circuits (2 courses)
- Electronics (2 courses)
- Computing Methods and Techniques for Engineering Analysis (C and MATLAB)
- Object Oriented Programming (C++)

Work Experience

Bangladesh Steel Re-Rolling Mill (BSRM)

January 2026 – Present

Engineer Electrical, Maintenance Electrical

BSRM is the leading company in steel industry in Bangladesh. During my industrial experience at BSRM, I encountered radio-based remote control systems used for transfer-trolley operation, Siemens SCALANCE industrial networking equipment used in Vibro Feeder operation, and infrared temperature measurement instruments for non-contact thermal sensing. These experiences helped me connect my academic background in electromagnetics and antenna engineering with practical applications involving RF communication, industrial automation and electromagnetic radiation-based sensing.

Research Interest

- Reconfigurable Antenna
- Electromagnetic Sensing
- Optical Sensing
- Microwave Photonics
- Computational Electromagnetics
- Biomedical Electromagnetics

Research Experience

Undergraduate Thesis:

Design of a Capacitor-based 2D variable Beam-Switching 2×2 Patch Array Antenna

This work presents the design and analysis of a 2D beam-shifting microstrip patch array antenna, aimed at generating a radiation pattern concentrated along the Z axis which can be steered along x-axis and y-axis. The proposed system integrates four rectangular patch antennas, arranged symmetrically and fed through a strip slot power divider, which ensures efficient and isolated power distribution. To achieve electronic beam steering, capacitors are incorporated within the transmission path, enabling phase tuning without mechanical movement.

Journal Paper(Work in Progress)

Design of a Varactor Diode-based 2D variable Beam-Switching 2×2 Patch Array Antenna

As I developed a design that shows beam-switching can be achieved with the change of capacitance, now design improvement is in progress. I am trying to achieve the capacitance change with variation in voltage level. For that target, we are using varactor diode.

Certificate

Digital Design for Industrial Control course

IICT, CUET and Bangladesh Computer Council

May 2025

This is an 80-hour course conducted by Institute of Information and Communication Technology (IICT), CUET under the project of Enhancing Digital Government and Economy, Digital Skills for Students, by Bangladesh Computer Council, ICT Division.

Python Data Structures

Coursera Certificate by University of Michigan

July 2020

Skills

Softwares: Igor pro, Overleaf, MS Office, MS Power Point, MS Excel, Google Doc

Programming Languages: C, C++, Matlab

Soft skills: Public speaking, Organizing, Documentation.

Leadership Skills

General Secretary

IEEE CUET Student Branch

Aug 2024 - Aug 2025

IEEE CUET Student Branch is a student chapter of the Institute of Electrical and Electronics Engineers (IEEE). IEEE CUET Student Branch is the largest technical organization in CUET consisting of almost 400 members with various activities for students.

Petitioner and Founding Chair

IEEE Photonics Society CUET Student Branch Chapter

Jan 2024 - Aug 2025

Webmaster

IEEE CUET Student Branch

July 2023 - August 2024

IEEE has a significant position as webmaster. Its primary duty is to report the Student Branch activity to the Global IEEE Vtools. My responsibility was to document all the activity of my organization and to report that activity to IEEE global.

Joint General Secretary

Collegiate School Debating Society (CSDS)

May 2015 - April 2016

Collegiate School Debating Society (CSDS) is a debating club. It was founded on 2014. Chittagong Collegiate School is one of the prominent high schools of Chattogram. CSDS is one of the best performing debate organization in Chattogram Debate Circuit.

Reference(s)

Dr. Mohammad Istiaque Reja

Professor, Department of Electrical and Electronic Engineering
Chittagong University of Engineering and Technology (CUET)

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Dr. Muhammad Asad Rahman

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